## (a) Early grouping decisions

Initial groups

r1: A
r2: A B
r3: D A B
r5: D C
r4: C

Planner

r3 alone:
reuse A B: +7
recompute D: -9

Group D:
D picked first
(saves 9)

r5 alone:
reuse C: +5
recompute D: -9

Each alone gains less than D's 9 tokens.

Planned prefill: 33 → 35 or 37 for a single move.

## (b) Costly regrouping checks

Many candidate moves
(who moves × which order)

Insert, read, undo
for every candidate.

Grows exponentially

repeat

Too many candidates to check one by one.

## (c) Prefix not ready

A in flight

**Admit too early**

r1: Compute A
r2: Compute A

Repeated work
time

**Wait in queue order**

r2: Compute B (A cached)
r3: Wait for A B (not ready)
r4: C queued

Independent work blocked

Admission must respect dependencies.

## (d) KaleidoKV (ours)

### 1. Joint regrouping

Move r3 and r5 together

r1: A
r2: A B
r3: A B D
r4: C
r5: C D

r3 and r5 leave D for different groups.

Planned prefill: 33 → 30.

Keep only improving moves.

Moves / cost change

### 2. Global prefix index

∅
r1 A 2 → 3 r4 C 1 → 2 D 2 → 0
r2 B 1 → 2 r5 D 0 → 1
r3 D 0 → 1
A C B

Positive → zero: subtract weight.

Zero → positive: add weight once.

Planned layout

### 3. Prefix-ready admission

**Nested Prefix**

A r1, r2, r3
A B r2, r3
A B D r3
C C D r5

**One level** — B computed twice

r1 A
r2 A B
r3 A B D

**Nested (ours)** — B computed once

r1 A
r2 A B
r3 A B D

Release level by level, run independent work.

Evidence documents (overlap across requests): A : 3 tokens B : 4 tokens C : 5 tokens D : 9 tokens A = computed (new prefill) A = reused (from cache)